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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B.Tech (CSE)
Name of the Paper: Python Programming

Semester: III
Paper Code: TCS346

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO2, CO4
(a)	List the different access modes of file in python? Compare between read(), readline(), readlines() methods? Demonstrate with the help of an example.	
(b)	You learnt about inheritance today in class but your friend has doubts regarding how it works. So you decided to explain him with a simple example of animal. Now, your task is to build single level inheritance with base class Animal and child class Dog. Illustrate the code snippet of class and methods.	
(c)	While using spotify you got an idea to create your own app. Being a programming enthusiast you started developing a dummy app first with basic functionalities. You need to design methods that will allow you to perform the following operations: a. Add the name of the song to the list. b. Remove the name of the song from the list. c. Display the names of the songs in sorted manner. d. Display the names of first 3 songs, last 3 songs (assuming you already have a list named playlist with 10 entries). Note: create only the methods.	
Q2	(20 marks)	CO1, CO5, CO3
(a)	Define five strings and list method, illustrate with code snippet.	
(b)	List at least 5 methods which are used for EDA. What is data cleaning and why it is important discuss a scenario.	
(c)	You are working as an intern for a gaming app. You have been assigned a task to design a feature for a puzzle game that checks if a number is a spy number or not (A spy number is a number where the sum and product of its digits is equal). Players tend to make errors, thus your manager wants you to do graceful handling of the scenarios by providing the user-friendly error messages. Develop a Python program that accepts the integer, check if it's a spy number or not and return an appropriate message to the user.	
Q3	(20 marks)	CO4, CO5
(a)	Discuss the term polymorphism? Compare method overloading and method overriding with the help of examples.	

(b)	You are working as a data analyst and your client wants you to analyze their stock price over the last one year and thus provide you with the month wise data of last one year stored as a numpy array having the month and the stock price. You have been asked to carry out the following analysis on the data: - Select the months having the minimum and maximum stock price. - Determine the mean and median of the data. - Evaluate the standard deviation and variance of the data. - Construct the line graph between month and stock price. - Reshape the array to visualize it as quarterly data.	
(c)	Rahul is a health conscious person, and since winter has started he decided to go for a walk every day and note down the following data Date, Day, Distance (km), Duration (hr). You have been provided with the data in .csv format. You are required to perform following tasks: - Read csv data as pandas dataframe. - List the days where Rahul covered more than 2.5 kms. - Find the dates when Rahul spent less than 1 hr for walking. - Update the record of distance column as 0.5 for the date 25/11/2024.	
Q4	(20 marks)	CO1, CO5, CO6
(a)	Summarize tuples in python? List the different ways to create a tuple in python? Demonstrate using a program how to perform various operations on tuple.	
(b)	Discuss the following with code snippet: - Box plot - Scatter plot - Histogram - Positive correlation - Negative correlation	
(c)	For your assignment you are required to collect the minimum temperature data over the period of one week over for the month of January and do analysis on that data. Being a python enthusiast you decided to create a program. Now your task is that the program should prompt user to input minimum temperature for seven days and store it in a numpy array, and perform following analysis: - Get the maximum and minimum temperature of the week. - List all the temperatures that are less than 7.5. - Sort the array in ascending order. - Find the position of top 2 minimum values.	
Q5	(20 marks)	CO2, CO5
(a)	Discuss the different data types in numpy? Explain 5 methods of numpy along with code snippet. How do you change the datatype of a numpy array?	
(b)	You have been provided with a dataset containing N features and M samples. You are asked to performed EDA on the data and show the results. Your task is to perform the following operations on the dataset: - Show the stats of the dataframe. - Show the top 10 entries from the dataframe. - Drop the values containing null values. - Find the number of unique entries in 'Name' column. - Check if dataset contains duplicates.	
(c)	Build a program for a hospital that has its data stored in a txt file with records as patientId, name, age, disease, status. - Write a method to read a file and display all its content. - Write a method to search and display the record of the patient based on the patientId entered by user. - Write a method to update the status as "Treatment completed" for the patientId entered by user.	